

3.13. Tensores no Espaço-Tempo Plano.

$$\begin{aligned} 3.1 \quad \tilde{W}(\vec{V}) &= W_\mu V^\mu = W_0 V^0 + W_1 V^1 + W_2 V^2 + W_3 V^3 \\ &= (-2) \cdot (3) + (1)(2) + (0)(1) + (0)(0) = -4 \end{aligned}$$

$$W_\mu V^\mu = -4$$

$$V_\mu = \eta_{\mu\nu} V^\nu$$

$$u=0 \rightarrow V_0 = \eta_{00} V^0 = (-1)(3) = -3$$

$$u=1 \rightarrow V_1 = \eta_{11} V^1 = (1)(2) = 2$$

$$u=2 \rightarrow V_2 = \eta_{22} V^2 = (1)(1) = 1$$

$$u=3 \rightarrow V_3 = \eta_{33} V^3 = (1)(0) = 0$$

$$V_\mu = (-3, 2, 1, 0)$$

$$W^\mu = \eta^{\mu\nu} W_\nu$$

De forma semelhante,

$$W^\mu = (2, 1, 0, 0)$$

3.2

$$T^{\mu\nu} A_\nu = B^\mu \quad \text{Vetor.} \quad (1,0)$$

$$T^\mu{}_\mu = C \quad \text{Escalar}$$

$$A_\mu B_\mu \quad \text{Mal escrito (índices iguais no mesmo lugar)}$$

$$R_{\mu\nu} A^\mu B^\nu = C \quad \text{Escalar}$$

$$A^\mu = B_\mu \quad \text{Mal escrito (são coisas diferentes)}$$

$$T^{\mu\nu} A_\mu B_\nu C^\rho = D^\rho \quad \text{Vetor} \quad (1,0)$$

X

3.3.

a) S $x^0 = t, \quad x^1 = x$

S' $x'^0 = t', \quad x'^1 = x'$

Por Galileu:

$$x'^0 = x^0, \quad \text{e} \quad x'^1 = x^1 + \alpha x^0$$

Inversa:

$$x^0 = x'^0, \quad \text{e} \quad x^1 = x'^1 - \alpha x'^0$$

Calculando o Jacobiano:

$$\frac{\partial x^\mu}{\partial x'^\nu}$$

$$\frac{\partial x^0}{\partial x'^0} = 1, \quad \frac{\partial x^0}{\partial x'^1} = 0$$

$$\frac{\partial x^1}{\partial x'^0} = \frac{\partial (x'^1 - \alpha x'^0)}{\partial x'^0} = -\alpha$$

$$\frac{\partial x^1}{\partial x'^1} = \frac{\partial (x'^1 - \alpha x'^0)}{\partial x'^1} = 1$$

$$\frac{\partial x'^\mu}{\partial x^\nu}$$

$$\frac{\partial x'^0}{\partial x^0} = 1, \quad \frac{\partial x'^0}{\partial x^1} = 0$$

$$\frac{\partial x'^1}{\partial x^0} = \alpha, \quad \frac{\partial x'^1}{\partial x^1} = 1$$

$$\frac{\partial x'^\mu}{\partial x^\nu} = \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix}$$

Assim:

$$\frac{\partial x^\mu}{\partial x'^\nu} = \begin{pmatrix} 1 & 0 \\ -\alpha & 1 \end{pmatrix}$$

Assim, para um vetor contravariante V^μ :

$$V'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} V^\nu$$

$$V'^0 = V^0, \quad e \quad V'^1 = \alpha V^0 + V^1$$

Para 1-form

$$\tilde{\omega}'_\nu = \frac{\partial x^\mu}{\partial x'^\nu} \tilde{\omega}_\mu$$

$$\tilde{\omega}'_0 = \tilde{\omega}_0 - \alpha \tilde{\omega}_1, \quad e \quad \tilde{\omega}'_1 = \tilde{\omega}_1$$

$$\begin{aligned} b) \quad \omega'_\mu V'^\mu &= \omega'_0 V'^0 + \omega'_1 V'^1 \\ &= (\omega_0 - \alpha \omega_1)(V^0) + (\omega_1)(\alpha V^0 + V^1) \\ &= \omega_0 V^0 - \cancel{\alpha \omega_1 V^0} + \cancel{\alpha \omega_1 V^0} + \omega_1 V^1 \\ &= \omega_0 V^0 + \omega_1 V^1 \end{aligned}$$

$$\omega'_\mu V'^\mu = \omega_\mu V^\mu \quad \text{X}$$

c) Invariância da Contração: Universal, não depende das coord. ou métricas.

Invariância da Métrica: Exige uma métrica!!!!

3.4.

a) $F^{\mu\nu} = -F^{\nu\mu}$ (Antissimétrica)

F^{ii} : $F^{00} = F^{11} = F^{22} = F^{33} = 0$

F^{0i} : $F^{0i} = E^i$, $F^{i0} = -E^i$

F^{ij} : $F^{ij} = e^{ijk} B_k$

Logo:

$$F^{12} = e^{12k} B_k = e^{123} B_3 = B^3, \quad F^{21} = -B^3$$

$$F^{23} = e^{23k} B_k = e^{231} B_1 = B^1, \quad F^{32} = -B^1$$

$$F^{31} = e^{31k} B_k = e^{312} B_2 = B^2, \quad F^{13} = -B^2$$

Logo:

$$F^{\mu\nu} = \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & B^3 & -B^2 \\ -E^2 & -B^3 & 0 & B^1 \\ -E^3 & B^2 & -B^1 & 0 \end{pmatrix}$$

b) $F_{\mu\nu} = \eta_{\mu\alpha} \eta_{\nu\beta} F^{\alpha\beta}$

Como a métrica é diagonal, só vão sobrar os termos em que $\mu = \alpha$ e $\nu = \beta$.

Assim:

$F_{0i} = -E^i$, $F_{i0} = E^i$, $F_{ij} = F^{ij}$ (não muda o sinal deste último).

Logo:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & B^1 & -B^2 \\ E^2 & -B^1 & 0 & B^3 \\ E^3 & B^2 & -B^3 & 0 \end{pmatrix}$$

Assim:

* Termos temporais

$$F^{00}F_{00} + \underbrace{F^{0i}F_{0i}}_{\text{somados}} + \underbrace{F^{i0}F_{i0}}_{\text{somados}} = -2\vec{E}^2$$

$\vec{E} = (E^1) + (E^2) + (E^3)$

* Termos espaciais

$$F^{12}F_{12} + F^{21}F_{21} + F^{23}F_{23} + F^{32}F_{32} + F^{31}F_{31} + F^{13}F_{13} =$$
$$= 2\vec{B}^2, \quad \vec{B} = (B^1) + (B^2) + (B^3)$$

Deste modo:

$$F^{\mu\nu}F_{\mu\nu} = 2(\vec{B}^2 - \vec{E}^2) \quad \text{UFA!!!}$$

c) Seja S um referencial onde:

$$\vec{B} = \vec{0} \text{ e } \vec{E} \neq \vec{0}$$

$$\text{Assim: } I = F^{\mu\nu}F_{\mu\nu} = 2(\vec{0}^2 - \vec{E}^2) = -2\vec{E}^2$$

Como $\vec{E} \neq 0$, $(\vec{E})^2 > 0$, logo $I < 0$.

Agora em S' , supomos:

$$\vec{E}' = \vec{0} \text{ e } \vec{B} \neq \vec{0}$$

$$\text{Assim: } I' = F^{\mu\nu}F_{\mu\nu} = 2(\vec{B}^2 - \vec{0}^2) = 2\vec{B}^2$$

Como $\vec{B} \neq 0$, $(\vec{B})^2 > 0$, logo $I' > 0$

ABSURDO! O escalar I é um invariante sobre qq transf. de Lorentz.

Não pode haver!!

3.5.

a) $T^{\mu\nu} = \rho U^\mu U^\nu + p h^{\mu\nu} + 2q^{(\mu} U^{\nu)} + \pi^{\mu\nu}$

$$T^{\mu\nu} = \rho U^\mu U^\nu + p h^{\mu\nu} + q^\mu U^\nu + q^\nu U^\mu + \pi^{\mu\nu}$$

P/p: (projetar $T^{\mu\nu}$ usando $U_\mu U_\nu$)

$$T^{\mu\nu} U_\mu U_\nu = \rho (\cancel{U^\mu U_\mu}) (\cancel{U^\nu U_\nu}) + p h^{\mu\nu} (\cancel{U_\mu}) (\cancel{U_\nu}) +$$

$$+ (q^\mu U^\nu + q^\nu U^\mu) U_\mu U_\nu + \pi^{\mu\nu} \cancel{U_\mu U_\nu}$$

$$\rho = T^{\mu\nu} U_\mu U_\nu$$

P/q (usando U_ν)

$$T^{\mu\nu} U_\nu = \rho U^\mu (\cancel{U^\nu U_\nu}) + p h^{\mu\nu} \cancel{U_\nu} + q^\mu (\cancel{U^\nu U_\nu}) + q^\nu \cancel{U_\nu} U^\mu + \pi^{\mu\nu} \cancel{U_\nu}$$

$$T^{\mu\nu} U_\nu = \rho U^\mu - q^\mu$$

$$T^{\mu\nu} U_\nu = -T^{\alpha\beta} U_\alpha U_\beta U^\mu - q^\mu$$

$$q^\mu = -T^{\mu\nu} U_\nu - (T^{\alpha\beta} U_\alpha U_\beta) U^\mu$$

Ou: $q^\mu = -h^\mu{}_\alpha T^{\alpha\nu} U_\nu$

P/p: (Usando $h^\alpha{}_\mu h^\beta{}_\nu$:

$$h^\alpha{}_\mu h^\beta{}_\nu T^{\mu\nu} = h^\alpha{}_\mu h^\beta{}_\nu (\rho U^\mu U^\nu + p h^{\mu\nu} + q^\mu U^\nu + q^\nu U^\mu + \pi^{\mu\nu})$$

Como: $h^\alpha{}_\mu U^\mu = 0$, temos:

$$h^\alpha{}_\mu h^\beta{}_\nu T^{\mu\nu} = h^\alpha{}_\mu h^\beta{}_\nu h^{\mu\nu} + h^\alpha{}_\mu h^\beta{}_\nu \pi^{\mu\nu}$$

Como: $h^\alpha{}_\mu h^{\mu\nu} = h^{\alpha\nu}$ e $h^\alpha{}_\mu \pi^{\mu\nu} = \pi^{\alpha\nu}$

Tem-se: $h^\alpha_\mu h^\beta_\nu T^{\mu\nu} = p h^{\alpha\beta} + \pi^{\alpha\beta}$

Tomando o traço (??)

$$\eta_{\alpha\beta} (h^\alpha_\mu h^\beta_\nu T^{\mu\nu}) = p h^\alpha_\alpha + \pi^\alpha_\alpha$$

Aqui:

$$\ast h^\alpha_\alpha = \cancel{4} + \cancel{U^\alpha_\alpha}^{-1} = 3$$

$$\ast \pi^\alpha_\alpha = 0 \quad (\text{enunciado})$$

$$\begin{aligned} \ast \eta_{\alpha\beta} (h^\alpha_\mu h^\beta_\nu T^{\mu\nu}) &= h^\nu_\nu T^{\mu\mu} = (\delta^\nu_\mu + U^\nu U_\mu) T^{\mu\nu} \\ &= T^\mu_\mu + \cancel{T^{\mu\nu} U_\mu U_\nu}_\rho = T^\mu_\mu + \rho \end{aligned}$$

Assim:

$$T^\mu_\mu + \rho = 3\rho$$

$$T^\mu_\mu + T^{\alpha\beta} U_\alpha U_\beta = 3\rho$$

Finalmente:

$$\rho = \frac{1}{3} (T^\mu_\mu + T^{\alpha\beta} U_\alpha U_\beta)$$

ρ está determinado somente pelas contrações de $T^{\mu\nu}$.

Para $\pi^{\mu\nu}$ (já calculamos antes)

$$\pi^{\alpha\beta} = h^\alpha_\mu h^\beta_\nu T^{\mu\nu} - \rho h^{\alpha\beta}$$

b) ρ - densidade local.

p - pressão isotrópica.

q^u - vetor de fluxo de calor.

π^{uv} - tensor de tensões de cisalhamento e viscosidade.

c) Lado esquerdo:

$T^{uv} \rightarrow 10$ (tensor simétrico 4×4).

Lado direito:

$\rho \rightarrow 1$ (escalar)

$p \rightarrow 1$ (escalar)

$q^u \rightarrow 3$ (4-vetor, porém $q^u U_u = 0$).

$\pi^{uv} \rightarrow 5$ (Tensor simétrico 4×4 , porém $\pi^{uv} U_v = 0$ (-4) e $\pi^u_u = 0$ (1))

$$(10 - 4 - 1) = 5$$

Assim: 10 graus de liberdade

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